

changed, allowing viability of satellite based systems—is untenable, since the corresponding cost structure of competing mobile systems has also significantly changed,” says Dr Borkar.

He explains that the large number of low-orbit satellites needed, with corresponding complexities and costs for management, make the business models of OneWeb and SpaceX somewhat questionable.

Other blocking factors include discrepancies in frequencies of operation, non-viability of real-time and interactive applications as well as high-bandwidth data applications due to limited capacity and bandwidth issues, and political differences and regulatory issues between different countries.

Dr Borkar adds, “These satellite based systems cannot compete with mobile systems, with their range of capabilities at different price points. This gives significant competitive advantages to different generations of mobile systems, from 2G to 5G. Clearly, there is a market for bicycles as well as cars in the telecommunications business.”

Dr Borkar’s strong faith in mobile systems is rooted in constant innovations, therein. Evolution of mobile systems, especially 2G Global System for Mobile (GSM) communication, 3G Wideband Code Division Multiple Access (WCDMA)/High-Speed Packet Access (HSPA), 4G LTE and 5G Next-Generation Networks (NGNs), are illustrations of continuous innovation for providing viable solutions to evolving market needs. Advances include Cooperative Multi-Point (CoMP) systems, multiple-input-multiple-output (MIMO) antennae, cloud based service creation and network management, and operations at extremely-high frequencies.

5G Wireless network is being positioned very well for meeting the advanced needs of the market in the 2020+ time frame, including mo-

bile multimedia and differentiated customer experience. These include ultra-high-definition (UHD) video, everything connected, sensor based M2M communications, smartcities, ubiquitous computing, real-time and cloud based services and instantaneous communication regardless of distance.

Coming to the question of how this helps in universal connectivity, Dr Borkar explains, “Mobile systems are scalable and can provide a range of viable options from low-cost voice services to extremely-high-bandwidth advanced services.

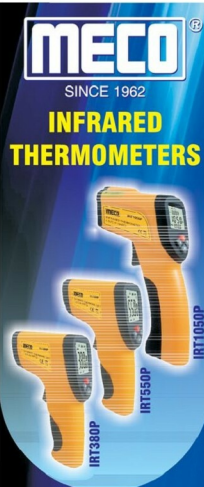
“5G NGN specifically is being designed to operate all the way from the lower frequency range of the 3G/4G spectrum in the 700MHz range to as high as 100GHz frequencies. Using a range of mobile systems including macrocells, microcells and picocells, it is well-positioned to provide selective as well as universal geographic coverage across the globe.

“In fact, due to advantages of seamless handovers, pico or femto cells are expected to be a better option than the currently ubiquitous Wi-Fi systems, especially for mobile devices. Wi-Fi has had its days, since nomadic access from devices like laptops without automatic handover to the mobile network has been the market norm.”

He adds, “There are applications where base stations based terrestrial networks are not feasible, for example, over the ocean, remote locations, high mountains and so on. For such niche markets, it is likely that SpaceX and OneWeb will combine and provide viable solutions.”

Eyes on television white-space

Another trending technology in this space is the utilisation of television white-space. From Google and Microsoft to indigenous companies like Saankhya Labs, and various universities including Indian Institutes of Technology (IITs), researchers are



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